



WMSU

HCI AND INTERACTION DESIGN



Objectives:

- Assess the relationship of Interactive Designs and HCI and apply the process and principles of interaction design.
- Assess the difference between good and poor interaction design.
- Demonstrate an understanding on what “Everyday Things” is
- Evaluate an Everyday Thing and explain what is good and bad about it in terms of the goals and principles of interaction design.



GOOD AND POOR DESIGN



Good and Poor Design

- A central concern of interaction design is to develop interactive products that are usable. By this is generally meant easy to learn, effective to use, and provide an enjoyable user experience.
- A good place to start thinking about how to design usable interactive products is to compare examples of well and poorly designed ones.



Example: Previous Activity

- How does making a phone call differ when using: a public phone box and a cell phone?
- How have these devices been designed to take into account :
 - (a) the kind of users?
 - (b) type of activity being supported?
 - (c) context of use?



COMPARISONS



KIND OF USERS

- **Public phones/phone box** are designed to be used by the general public. Many have Braille embossed on the keys and speaker volume control to enable people who are blind and hard of hearing to use them.
- **Cell phones** are intended for all user groups, although they can be difficult to use for people who are blind or have limited manual dexterity.



Activity being supported

- Most phone boxes are designed with a simple mode of interaction: insert card or money and key in the phone number. If engaged or unable to connect the money or card is returned when the receiver is replaced.
- There is also the option of allowing the caller to make a follow-on call by pressing a button rather than collecting the money and reinserting it again. This function enables the making of multiple calls to be more efficient.



Activity being supported

- Cell phones have a more complex mode of interaction. More functionality is provided, requiring the user to spend time learning how to use them. For example, users can save phone numbers in an address book and then assign these to "hotkeys," allowing them to be called simply through pressing one or two keys.



Context of Use

- Phone boxes are intended to be used in public places, say on the street or in a bus station, and so have been designed to give the user a degree of privacy and noise protection through the use of hoods and booths.
- Cell phones have been designed to be used any place and any time. However, little consideration has been given to how such flexibility affects others who may be in the same public place (e.g., sitting on trains and buses).



What to design?

- Designing usable interactive products thus requires considering **who is going to be using them** and **where they are going to be used**.
- **Understand the kind of activities** people are doing when **interacting** with the products.



What to design?

- The range of activities that can be supported is **diverse**.

The top of the slide features a red banner. On the left, a white university seal is partially visible, showing an eagle and the word 'UNIVERSITY'. On the right, a photograph of a large, multi-story yellow building with arched windows and green trees in front of it is visible.

Question ?????

How do you optimize the users' interactions with a system, environment or product, so that they match the users' activities that are being supported and extended?



ANSWER:

- Take into account what people are good and bad at
- Consider what might help people with the way they currently do things
- Think through what might provide quality user experiences
- Listen to what people want and getting them involved in the design
- Use "tried and tested" user-based techniques during the design process



DONALD A. NORMAN ON EVERYDAY THINGS



Donald A. Norman

- Cognitive psychologist
- University of California San Diego
- The Psychology of Everyday Things
- Things that Make Us Smart
- The Invisible Computer
- Concerned with good design
- Fitting things to people, not the other way around
- Formulated principles of good design



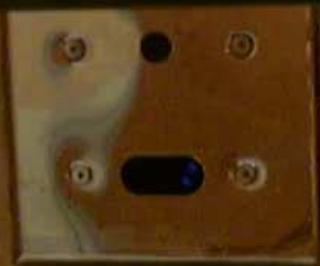
DESIGN PRINCIPLES



1. VISIBILITY

- Design makes the conceptual model apparent to users
- Design tells the users what actions they can perform
- Getting trapped between glass doors
- Hinge was not obvious
- No handles, no panels

FOR MANUAL FLUSHING
PLEASE PRESS THE
BLACK BUTTON



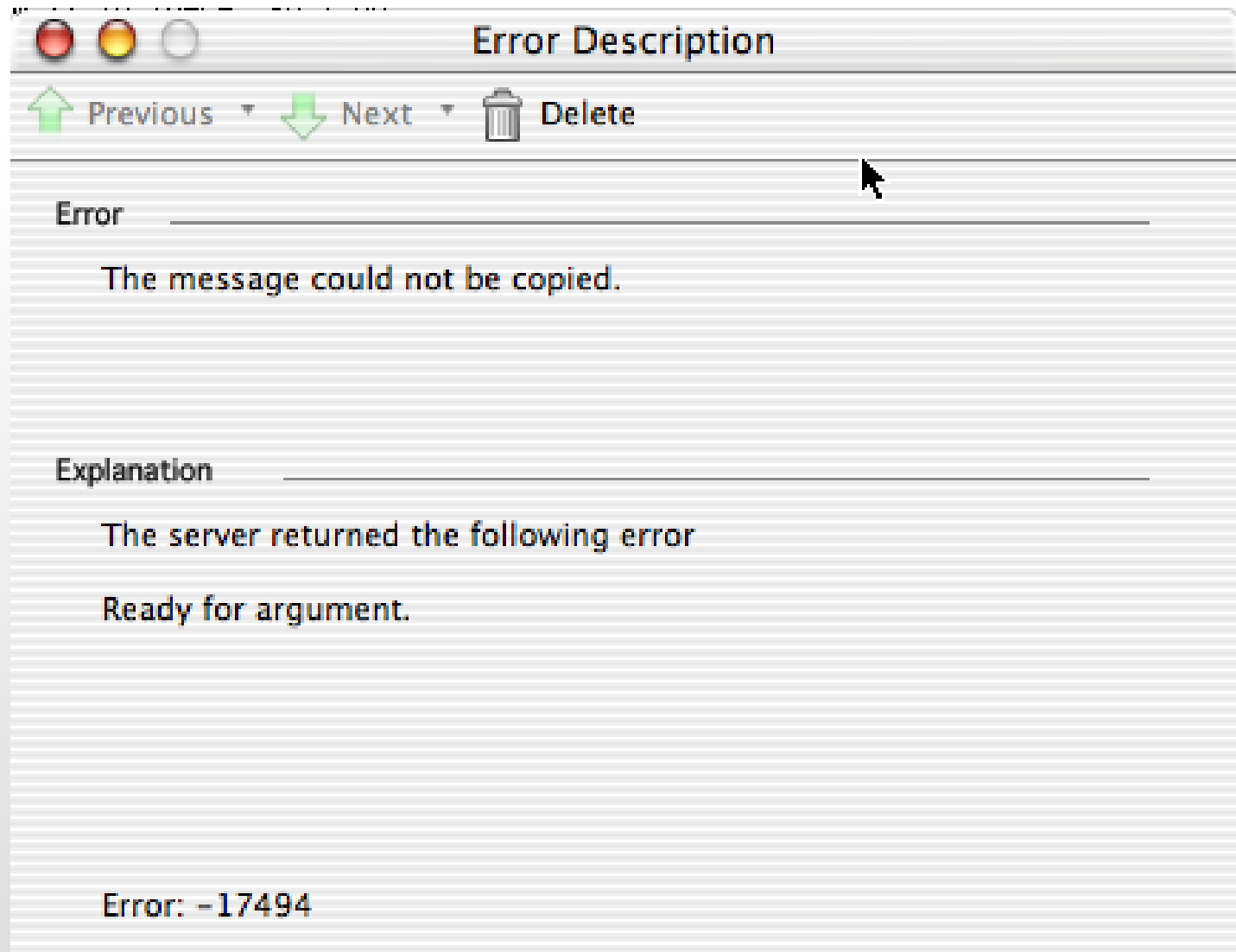


2. FEEDBACK

- Provides information about the effects of a user's actions
- What has been done
- What has been accomplished
- Auditory feedback (sidetone) when talking on a telephone
- Tones when buttons are pressed









3. MAPPING

- Relationships between controls and their effects on a system







4. AFFORDANCE

- Perceived properties of an artifact
- Provides strong clues for possible usage





5. CONSISTENCY

- This refers to designing interfaces to have similar operations and use similar elements for achieving similar tasks.



6. CONSTRAINTS

- The design concept of constraining refers to determining ways of restricting the kind of user interaction that can take place at a given moment.
- It can be classified:
 - Physical Constraints
 - Semantic Constraints
 - Cultural Constraints
 - Logical Constraints



DESIGN CONSTRAINTS



Physical Constraints

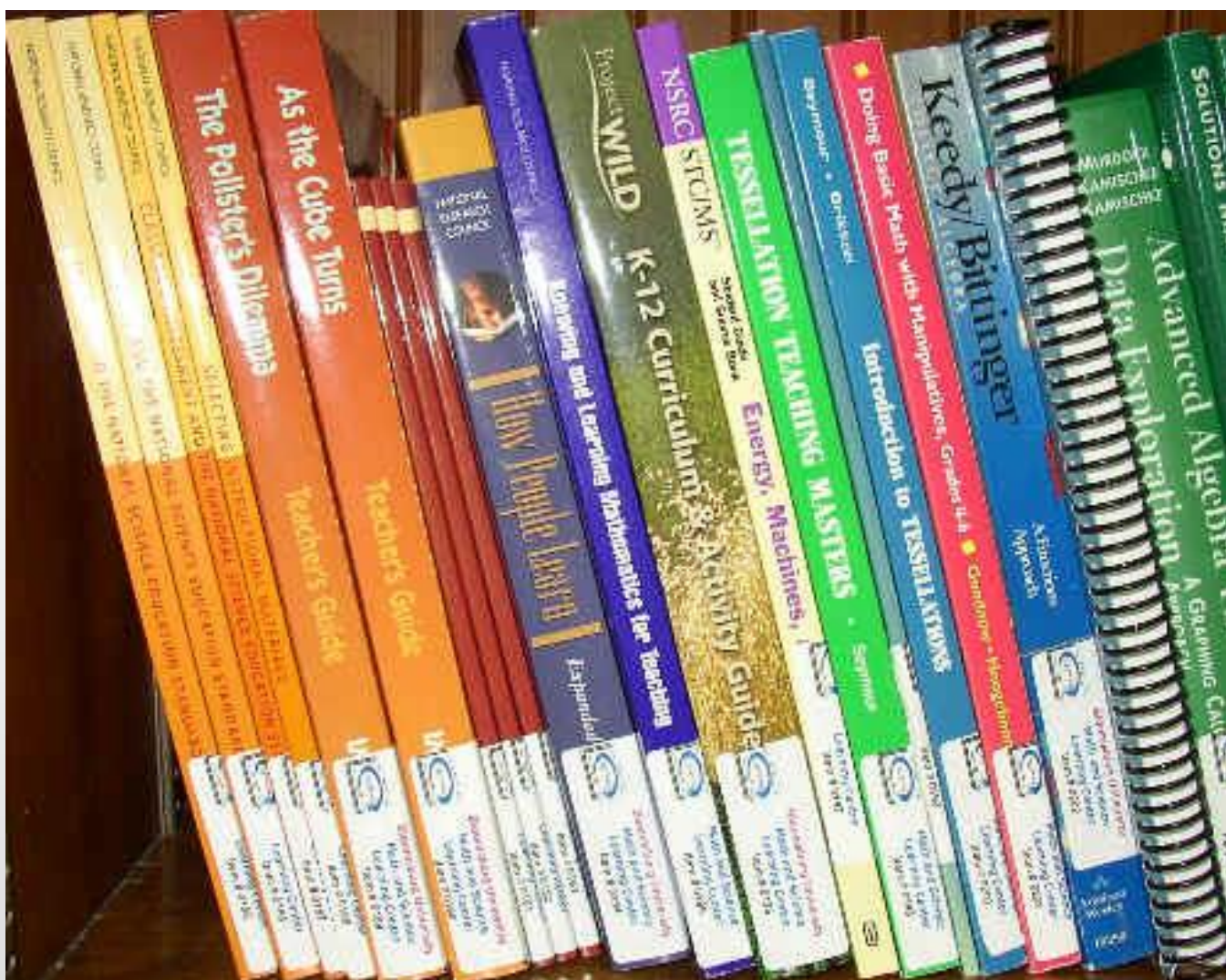
- Constrain possible operations
- refer to the way physical objects restrict the movement of things.
- Examples
 - different keys fit different locks





Semantic Constraints

- Rely on the meaning of a situation to control actions

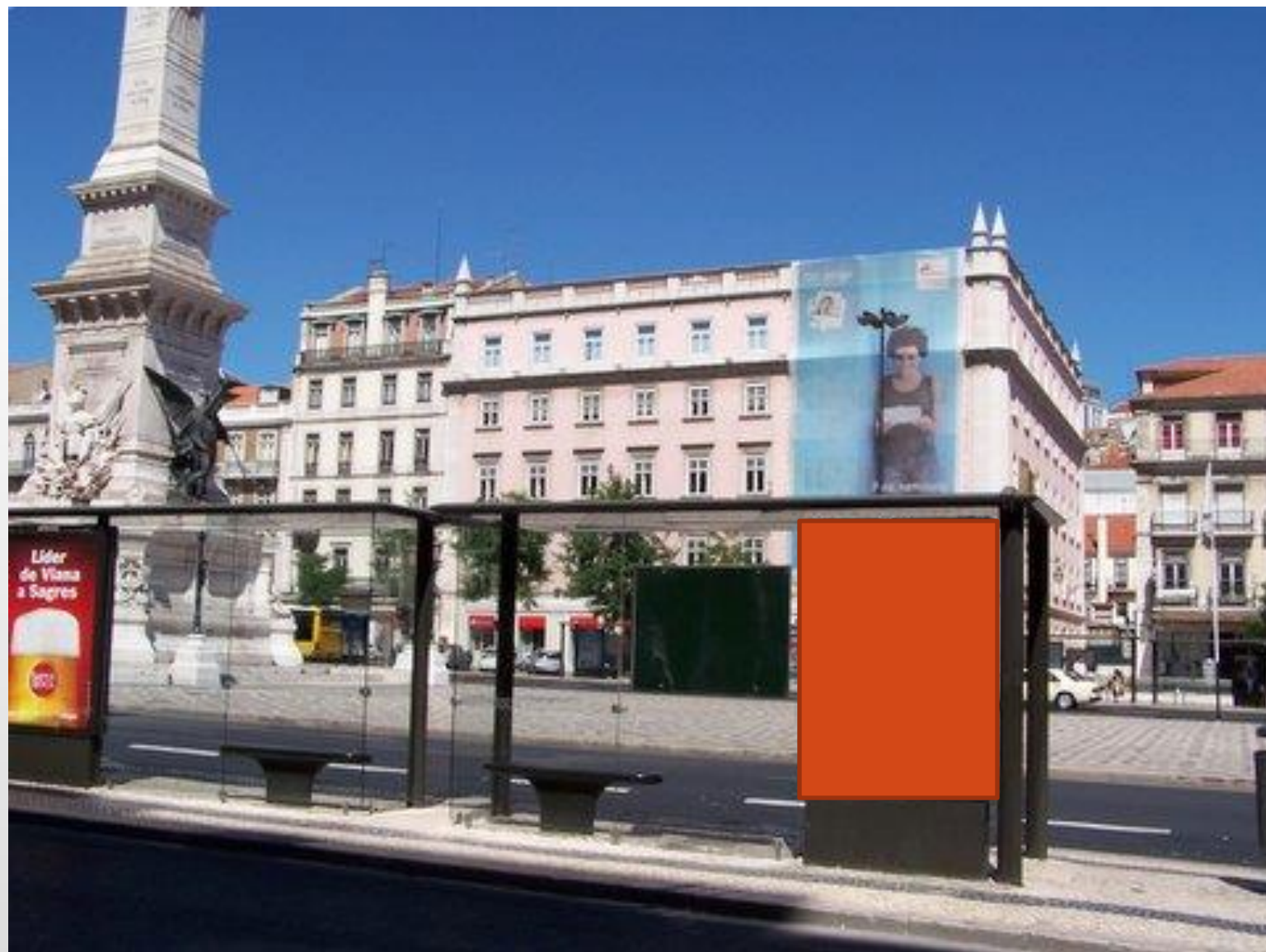






Cultural Constraints

- rely on learned convention
- Accepted cultural conventions that control actions







Logical Constraints

- rely on people's understanding of the way the world works
- They rely on people's common-sense reasoning about actions and their consequences.
- Relationship between spatial or functional layout of components and the things they affect





WHAT' S WRONG WITH THIS PICTURE?





















ACTIVITY: TAKE HOME



Activity #2: Everyday Thing

- Choose an everyday thing, preferably something that is not computer or information technology-based and analyze it based on the principles discussed in class. This everyday thing is something you have personally seen, used, experienced, or heard of (through friends, etc.). Important: **This analysis should NOT be taken from the web.**



Guide questions include

What is the object's function?

How do you use/operate the object?

What is odd or unusual about its design?

What principles did it the object's design follow?

Which principles did the design violate?

Do these design quirks make it more or less usable?

- Submissions must be one to two pages long, double-space, 12-point font with one-inch margins on all sides. Submission must include a picture of the everyday thing. The picture is included in the page count.
- **This is to be submitted on Tuesday at 1:30 to 2pm
Only Faculty Room**



INTERACTION DESIGN



What is Interaction Design?

- **“Designing interactive products to support people in their everyday and working lives.”**
- **“The design of spaces for human communication and interaction.” - Winograd (1997)**



The makeup of interaction design

- The importance of understanding how users act and react to events and how they communicate and interact together has led people from a variety of disciplines, such as psychologists and sociologists, to become involved.



The makeup of interaction design

- the growing importance of understanding how to design different kinds of interactive media in effective and aesthetically pleasing ways has led to a diversity of other practitioners becoming involved, including graphic designers, artists, animators, photographers, film experts, and product designers



Brief History of Interaction Design

- In the early days, engineers designed hardware systems for engineers to use.
- The computer interface was relatively straightforward, comprising various switch panels and dials that controlled a set of internal registers. With the advent of monitors (then referred to as visual display units or VDUs) and personal workstations in the late '70s and early '80s, interface design came into being (Grudin, 1990).



Brief History of Interaction Design

- One of the biggest challenges at that time was to develop computers that could be accessible and usable by other people, besides engineers, to support tasks involving human cognition (e.g., doing sums, writing documents, managing accounts, drawing plans).



Brief History of Interaction Design

- Computer scientists and software engineers developed high-level programming languages (e.g., BASIC, Prolog), system architectures, software design methods, and command-based languages to help in such tasks, while psychologists provided information about human capabilities (e.g., memory, decision making).



Brief History of Interaction Design

- Research into and development of graphical user interfaces (GUI for short, pronounced "goo-ee") for office-based systems took off in a big way.



Brief History of Interaction Design

- In the mid '80s, the next wave of computing technologies such as speech recognition, multimedia, information visualization, and virtual reality-presented even more opportunities for designing applications to support even more people.



Brief History of Interaction Design

- As further waves of technological development surfaced in the '90s-networking, mobile computing, and infrared sensing-the creation of a diversity of applications for all people became a real possibility.



Brief History of Interaction Design

- In the mid '90s, many companies realized it was necessary again to extend their existing multidisciplinary design teams to include professionals trained in media and design, including graphical design, industrial design, film, and narrative.

Brief History of Interaction Design

- Now in the 'OOs, the possibilities afforded by emerging hardware capabilities-e.g., radio-frequency tags, large interactive screens, and information appliances has led to a further realization that engineers, who know about hardware, software, and electronics are needed to configure, assemble, and program the consumer electronics and other devices to be able to communicate with each other (often referred to as middleware).



What is involved in the process of interaction design?

- Essentially, the process of interaction design involves four basic activities:
 1. Identifying needs and establishing requirements.
 2. Developing alternative designs that meet those requirements.
 3. Building interactive versions of the designs so that they can be communicated and assessed.
 4. Evaluating what is being built throughout the process.



Three key characteristics of the interaction design process:

- 1. Users should be involved through the development of the project.
- 2. Specific usability and user experience goals should be identified, clearly documented, and agreed upon at the beginning of the project.
- 3. Iteration through the four activities is inevitable.



The goals of interaction design

1. Usability

- effective to use (effectiveness)
- efficient to use (efficiency)
- safe to use (safety)
- have good utility (utility)
- easy to learn (learnability)
- easy to remember how to use (memorability)

- **Efficiency** refers to the way a system supports users in carrying out their tasks.
- **Safety** involves protecting the user from dangerous conditions and undesirable situations. It also refers to the perceived fears users might have of the consequences of making errors and how this affects their behavior.
- **Utility** refers to the extent to which the system provides the right kind of functionality so that users can do what they need or want to do.
- **Learnability** refers to how easy a system is to learn to use.
- **Memorability** refers to how easy a system is to remember how to use, once learned. This is especially important for interactive systems that are used infrequently. If users haven't used a system or an operation for a few months or longer, they should be able to remember or at least rapidly be reminded how to use it.



The goals of interaction design

2 . User Experience

creating systems that are:

- satisfying
- enjoyable
- fun
- entertaining
- helpful
- motivating
- aesthetically pleasing
- supportive of creativity
- rewarding
- emotionally fulfilling



Heuristics and usability principles

- Heuristics Principles
 - emphasizes that something has to be done with them when they are applied to a given problem.
 - Design principles are being practiced
- Usability Principles
 - "speak the user's language."



Ten Usability principles

- Developed by Nielsen(2001) and his colleagues
- 1. **Visibility of system status**-always keep users informed about what is going on, through providing appropriate feedback within reasonable time
- 2. **Match between system and the real world**-speak the users' language, using words, phrases and concepts familiar to the user, rather than system oriented terms



Ten Usability principles

3. **User control and freedom**- provide ways of allowing users to easily escape from places they unexpectedly find themselves, by using clearly marked 'emergency exits'
4. **Consistency and standards** -avoid making users wonder whether different words, situations, or actions mean the same thing



Ten Usability principles

- 5. Help users recognize, diagnose, and recover from errors-use plain language to describe the nature of the problem and suggest a way of solving it**
- 6. error prevention-where possible prevent errors occurring in the first place**
- 7. Recognition rather than recall-make objects, actions, and options visible**



Ten Usability principles

8. **Flexibility and efficiency of use** - provide accelerators that are invisible to novice users, but allow more experienced users to carry out tasks more quickly
9. **Aesthetic and minimalist design** - avoid using information that is irrelevant or rarely needed
10. **Help and documentation** - provide information that can be easily searched and provides help in a set of concrete steps that can easily be followed



Key Points

- Interaction design is concerned with designing interactive products to support people in their everyday and working lives.
- Interaction design is multidisciplinary, involving many inputs from wide-reaching disciplines and fields.
- Interaction design is now big business: many companies want it but don't know how to do it.
- Optimizing the interaction between users and interactive products requires taking into account a number of interdependent factors, including context of use, type of task, and kind of user.



Key Points

- Interactive products need to be designed to match usability goals like ease of use and learning.
- User experience goals are concerned with creating systems that enhance the user experience in terms of making it enjoyable, fun, helpful, motivating, and pleasurable.
- Design and usability principles, like feedback and simplicity, are useful heuristics for analyzing and evaluating aspects of an interactive product.



ACTIVITY

Objectives :

1. To enable you to define usability and user experience goals
1. To use design and usability principles for evaluating the usability of an interactive product.



ACTIVITY #3:

- Group yourselves into three
- Do the following:

Find a handheld device (e.g. remote control, handheld computer, or cell phone) and examine how it has been designed, paying particular attention to how the user is meant to interact with it.



Guide Questions:

- (a) From your first impressions, write down what first comes to mind as to what is good and bad about the way the device works. Then list (i) its functionality and (ii) the range of tasks a typical user would want to do using it. Is the functionality greater, equal, or less than what the user wants to do?



Guide Questions:

(b) Based on your reading of this chapter and any other material you have come across, compile your own set of usability and user experience goals that you think will be most useful in evaluating the device. Decide which are the most important ones and explain why.



Guide Questions:

(c) Translate the core usability and user experience goals you have selected into two or three questions. Then use them to assess how well your device fares (e.g., Usability goals. What specific mechanisms have been used to ensure safety? How easy is it to learn? User experience goals: Is it fun to use? Does the user get frustrated easily? If so, why?).



Guide Questions:

- (d) Repeat (b) and (c) for design concepts and usability principles (again choose a relevant set).
- (e) Finally, discuss possible improvements to the interface based on your usability evaluation.